Chapter 1

Objectives

- Explain the difference between Computer system and network administration
- Understand Responsibilities & goals of the System Administrator
- List different types of network administrators
- Understand The Philosophy of System Administration
- List different Network Operating System Software

INTRODUCTION

What is 'the system'?

In system administration, the word ***system*** is used to refer both to the **operating system of a computer** and often, collectively the **set of all computers that cooperate in a network.**

If we look at computer systems analytically, we will speak more precisely about ***human–computer systems***.

human–computer system: An *organized collaboration between humans and computers to solve a problem or provide a service.* Main components in HCS:

Humans, Host computers, & Network hardware

What is network and system administration?

- System and network administration is a branch of engineering that concerns the operational management of human computer systems.
- System administration is not just an administrative job; it is an extremely demanding engineer's job.
- It's about hardware, software, user support, diagnosis, repair and prevention.
- It is about putting together a network of computers (workstations, PCs etc.), getting them running and then keeping them running despite the activities of users who tend to cause the systems to fail.

Cont.

- **System administration** is the field of work in which someone manages one or more systems that is either software, hardware, servers or workstations.
- **System administration** is the term used traditionally by mainframe and Unix engineers to describe the management of computers whether they are coupled by a network or not.
- Now a days **network administration** means the management of network infrastructure devices (routers and switches).

Cont.

- **A system administrator (Sys Admin)** is a person responsible to maintain and operate a computer system or network for a company or other organization.
- **Sys Admin** has the knowledge of **operating systems** and **applications**, as well as **hardware and software** troubleshooting.
- System administrators need to know a bit of everything and have technical, administrative and socio-psychological skills.
- A system administrator works for users. However, a system administrator should not just provide for one or two selfish needs but also work for the benefit of a whole community.

Responsibilities of the System Administrator

- Add and remove users.
- Add and remove hardware.
- Perform backups.
- Install new software systems.
- Troubleshooting.
- System monitoring.
- Auditing security.
- Helping users.

Goals of System/Network Administration

- Ensure that computing systems run correctly and as efficiently as possible
- Ensure that users can and do use the computing systems to carry out their required work in the easiest and most efficient manner.

What is special about the system administrator account?

System administrator has **full (unrestricted) access** resources and performing different tasks, so he/she can do anything with system. For example, system administrator can remove critical system files.

Some of the tasks accomplished by system administrators are :

- Create new users
- Resetting user passwords
- Lock/unlock user accounts
- Monitor server security
- Monitor special services etc.

Types of Administrators/Users

- In a larger company, there are different system administrators in separate positions within a computer support or Information Services (IS) department.
- But in a smaller company or organization, all tasks can be administrated by a few system administrator, or even a single person.

Cont.

- **Database administrator** (integrity of the data)
- **Network administrator** (network infrastructure)
- **Security administrator**(security devices)
- **Web administrator** (maintains web server services)
- **Computer operator (Technical support)** performs routine maintenance.

The challenges of system administration

Some of the challenges of system administrators are the following :

- Designing a network which is efficient.
- Deploying large numbers of machines which can be easily upgraded later.
- Deciding what services are needed.
- Implementing adequate security.
- Developing ways of fixing errors

As a job, Systems Administration is

- complex
- painful
- annoying
- highly interactive
- well paid
- employable and thankless

The Philosophy of System Administration

Although the specifics of being a system administrator may change from platform to platform, there are underlying themes that do not. These themes make up the philosophy of system administration.

The themes are:

- Automate everything
- Document everything
- Communicate as much as possible
- Know your resources
- Know your users
- Know your business
- Make the systems Secure
- Plan ahead
- Expect the unexpected

Automate Everything

Anything done more than once should be automated in order to save our time, energy, and replace human errors.

Here are some commonly automated tasks:

- Free disk space checking and reporting
- Backups
- System performance data collection
- Business-specific functions (pushing new data to a Web server, running monthly/quarterly/yearly reports, etc.)

Document Everything

- Policies
- Procedures
- Changes

Otherwise, you could find yourself being completely confused about a change you made several months earlier.

Communicate as Much as Possible

Some organizations use email; others, an internal website. In any case, use whatever method(s) that work well at your organization.

- Tell your users what you are going to do.
- Tell your users what you are doing.
- Tell your users what you have done.

Know your Resources

System administration is mostly a matter of **balancing available resources against the people** and **programs that use those resources**.

- System resources, such as **available processing power, memory, and disk space**
- Network bandwidth
- Available money in the IT budget

Know your Users

- Users are those people that use the systems and resources for which you are responsible.
- Identify power users from others.
- Make sure you learn as much about your users as you can.

Know Your Business

- Whether you work for a large, multinational corporation or a small community college, you must still understand the nature of the business environment in which you work.
- This can be boiled down to one question: i.e. **What is the purpose of the systems you administer?**

Make the systems Secure

The following list illustrates the different kinds of issues you should consider:

- The nature of possible threats to each of the systems under your care.
- The location, type, and value of the data on those systems.
- The type and frequency of authorized access to the systems

Plan Ahead

- Certainly, no one can predict the future with 100% accuracy.
- However, with a bit of awareness it is easy to read the signs of many changes in the future.
- So system administrator should consider future developments or changes when you install networks or other resources.

Expect the Unexpected

- What would a system administrator who expects the unexpected do in this case? Perhaps it is possible to keep a few disk drives sitting on the shelf as spares in case of hardware problems.
- By trying to anticipate problems before they occur, you will be in a position to respond more quickly and effectively than if you let yourself be surprised.

Network Operating System Software

Network operating system refers to a software provides the mechanism to integrate all the components on a network to allow multiple users to simultaneously share the same resources regardless of physical location. For example, to manage **data, users, groups, security, applications,** and other networking functions like **provide services on shared file & printer**.

The following are network operating systems software.

- Macintosh OS X,
- Microsoft Windows Server,
- UNIX/Linux

Microsoft Windows Operating System

The Windows is a operating system. This allows people to **manage files** and **run software programs** on desktop and laptop computers.

The Windows operating system is developed and maintained by **Microsoft**, the company founded by Bill Gates. The first version was released in 1985. Since then, it has grown significantly.

Windows uses a graphical user interface (**GUI**) to make it easier for people to use them. The primary ways that people navigate are through **icons** on the desktop and the **Start menu.** E.g. Win 98, Win 2000, Win Me, Win 2003,Win XP, Win Vista etc. All Flavors of Windows Come from Microsoft.

Linux/Unix

- Linux stands for Linus' Unix **which** originally developed by Linus Torwalds, Linux built upon the **Linux Kernel**(Unix tradition). Those kernel makes sure everything works **smoothly** and **efficiently**. But the Linux Kernel alone is not enough to make a complete operating system, the Linux Kernel is combined with a **collection of software packages and utilities**, which are together called **Linux distributions**. Also known as **Linux Operating System.** Linux distributions come in different form to suit the specific needs and preferences of users. Some popular Linux distributions are:
- Ubuntu, Debian, Fedora, CentOS, & Red Hat Enterprise Linux

Macintosh OS

- MacOS is a Unix-based operating system designed fully for Apple's Macintosh (Mac) computers. The **reliability**, **security** features, **ease of use**, and **compatibility** with other Apple goods and services are all well known aspects of macOS. It include a number of built in programs are for a variety of functions, productivity, email, multimedia playing, and web browsing.
- MacOS is characterized by a **user-friendly graphical user interface** (**GUI**), traditionally design **single button moussing**, and **nearly universal plug-and-play support**.
- In the spring of 2001, Apple transitioned from its original code base to MacOS X, a very different operating system based on **Unix** and the **Mach microkernel**. Like its predecessor, MacOS X maintains the traditional MacOS **ease of use**, but with substantially **improved stability**

Questions ?

Chapter 2

Objectives

- To understand Windows Network Concepts
- To Define Workgroup and Server Domain
- To Explain the difference between Workgroup and Server Domain
- To List advantage and disadvantage of Workgroup and Server Domain
- To Describe Active Directory
- To Define Forest, Tree, and Domain
- To Explain the logical and physical structure of Active Directory
- To List Trust relationship between different domains
- To describe what is DHCP & It's advantage
- To describe DNS Server

Workgroup

- A workgroup is a grouping of computers that are connected to each other over a network.
- In computer networking, a workgroup is a collection of computers on a local area network (LAN) that share common resources and responsibilities.
- Workgroups provide easy sharing of files, printers and other network resources.
- Workgroups are designed for small LANs in homes, schools, and small businesses.
- Workgroup functions best with 15 or fewer computers.

Cont.

- This grouping is handled within the Microsoft Windows operating system, where the members of the workgroup assume the same workgroup name (though each computer in a workgroup must have a unique computer name).
- Computers in a workgroup communicate directly with each other and do not require a server to manage network resources.
- Being a peer-to-peer (P2P) network design, each workgroup computer may both share and access resources if configured to do so.

Adding a computer to your workgroup

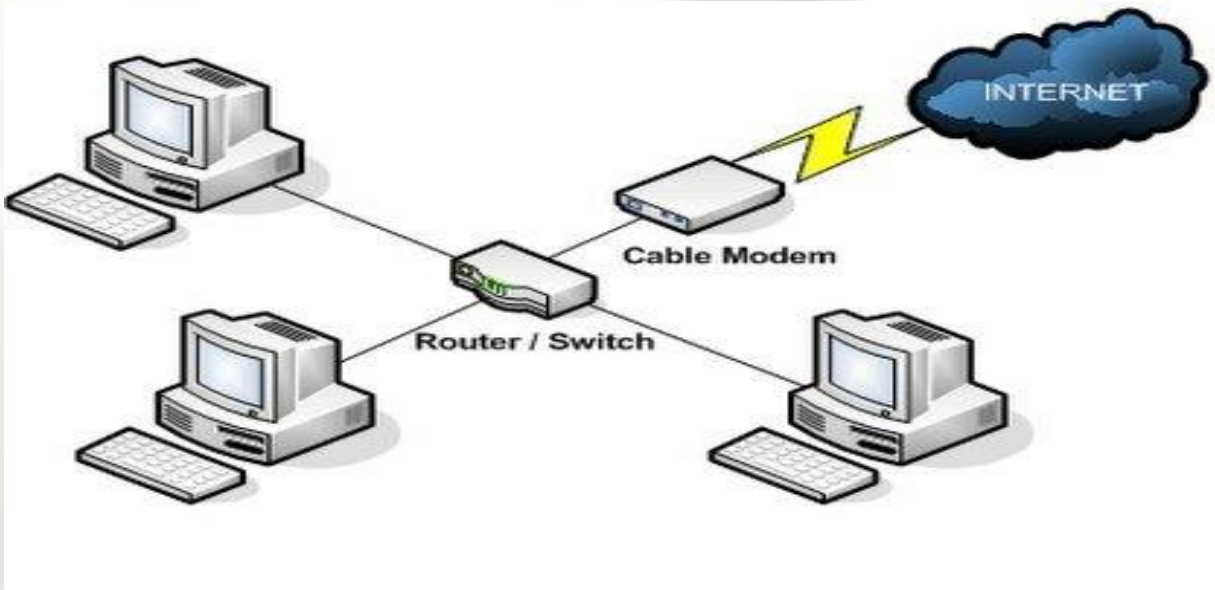
- After your network and workgroup are set up, you can add computers to the workgroup.
- The procedure varies slightly for different Microsoft Windows operating systems.
- **Be sure to enter the same workgroup names on all computers you are adding to the workgroup,** so that they appear together in My Network Places.

Cont.

Once a workgroup has been created, it is visible in My Network Places (available from the desktop in Windows).

As the number of computers in a workgroup grows, workgroup LANs eventually become too difficult to administer and should be replaced with alternative solutions like domains or other client/server approaches.

Fig: workgroup computers



Server Domain

A **domain** is defined as a logical group of network objects (computers, users, devices) that share the same active directory database.

- A domain is a management boundary.
- Domains are part of a **forest**.
- The first domain in a forest is known as the **forest root domain**.
- Domains, workgroups, and home groups represent different methods for organizing computers in networks.
- The main difference among them is how the computers and other resources on the networks are managed.

In a workgroup:

- All computers are peers; no computer has control over another computer.
- Each computer has a set of user accounts.
- To log on to any computer in the workgroup, you must have an account on that computer.
- There are typically no more than twenty computers.
- A workgroup is not protected by a password.
- A homegroup is protected with a password, but you only need to type the password once, when you are adding your computer to the homegroup.
- All computers must be on the same local network or subnet.

In a domain:

- One or more computers are servers.
- Network administrators use servers to control the security and permissions for all computers on the domain.
- This makes it easy to make changes because the changes are automatically made to all computers.
- Domain users must provide a password or other credentials each time they access the domain.
- If you have a user account on the domain, you can log on to any computer on the domain without needing an account on that computer.
- There can be thousands of computers in a domain.
- The computers can be on different local networks.

Disadvantages of a Domain

- The major, disadvantage of having a domain oriented network environment is that **it has a single point of failure.**
- This means that if the domain controller fails because of any reason, the entire network goes down, and comes up only when the domain controller starts working properly again.

What is the Active Directory (AD)

- An active directory is a **directory structure** used on Microsoft Windows based computers and servers **to store information and data about networks and domains.**
- It is primarily used for online information and was originally created in 1996 and first used with Windows 2000.

An active directory does a variety of functions including

- ❑The ability to provide information on objects,
- ❑Helps to organize the objects for easy retrieval and access
- ❑Allows access by end users and administrators
- ❑Allows the administrator to set up security for the directory.

Cont.

- In fact, an active directory can be viewed at either one of three levels, these levels are called forests, trees or domains.
- The highest structure is called the forest because you can see all objects included within the active directory.

An active directory can be defined as a hierarchical framework of objects. The objects fall into three main categories,

✓ **Resources** which may include hardware such as printers,

✓ **Services** for end users such as web email servers and

✓ **Objects** which are the main functions of the domain and network.

Cont.

- Remember that an object can be a piece of hardware such as a printer, end user or security settings set by the administrator.
- These objects can hold other objects within their file structure.
- Every object has its own attributes which allows it to be characterized by the information which it contains.
- For instance, some objects with certain schemas can not be deleted, they can only be deactivated.
- Others types of schemas with certain attributes can be deleted entirely.

Example of AD objects: **Computers, Groups, & Users**

Cont.

Some of the common tasks accomplished with Active Directory Users and Computers include:

- Adding new users to Active Directory
- Changing passwords
- Granting rights to file servers
- Allowing remote access to the network
- Setting login and logout scripts
- Restrict logon times
- Controlling when users can use the network
- Creating security groups - with either static or dynamic membership

Need of Active Directory Domain Services (AD DS)

- In medium to large scale industries where there are several computers in the network, it becomes practically impossible for the administrators to manage each computer individually.
- In such scenarios, **administrators** can install Active Directory Domain Services **on Windows Server 2012 R2** in order to centrally manage all computers in the network simultaneously, with least administrative overhead, and right from their own desks.

Cont.

When Active Directory Domain Services is installed on Windows Server 2012 R2, the computer is technically known as a **Domain Controller** , which then becomes capable of managing, authenticating, and providing services **to all client computers** that are added as its members.

- After the successful installation of Active Directory Domain Services, administrators can create domain user accounts right on the domain controller.
- Domain user accounts can log on to the domain using any of the available domain client computers.

Structure of Active directory

Structure of active directory is divided in two categories

- Logical structure
- Physical structure

Logical structure of AD

- **Logical structure** of AD includes **forest, tree, domains and organizational unit.**
- Objects are grouped into domains.
- The objects for a single domain are stored in a single database (which can be replicated).
- Domains are identified by their DNS name structure.

I) Domain

- A **domain** is defined as a logical group of network objects (computers, users, devices) that share the same active directory database.
- A domain is a management boundary.
- Domains are part of a **forest**.
- The first domain in a forest is known as the **forest root domain**.
- In many small and medium organizations (and even some large ones), you will only find a single domain in a single forest.

Cont.

- The forest root domain defines the default namespace for the forest.
- For example, if the first domain in a new forest is named **domain1.com**, then that is the forest root domain. If you have a business need for a child domain, for example - a branch office in Gondar, you might name the child domain chi.
- The child domain would be **chi.domain1.com.**

II) Tree

- A **tree** is a collection of one or more domains.

III) Forest

- At the top of the structure there is the forest.
- **A forest** is a collection of trees that **share** a common global catalog, directory schema, logical structure, and directory configuration.
- The forest represents the **security boundary** within which users, computers, groups, and other objects are accessible.

Cont.

- Objects in separate forests are not able to interact with each other, **unless** the administrators of each separate forest create a **trust** between them.
- For example, an Enterprise Administrator account for domain1.com, which is normally the most privileged account of a forest, will have, no permissions at all in a second forest named domain2.com, even if those forests exist within the same LAN, unless there is a trust in place.

IV) Organizational units

- The objects held within a domain can be grouped into Organizational Units (OUs).
- OUs can provide hierarchy to a domain, ease its administration, and can resemble the organization's structure in managerial or geographical terms.
- OUs can contain other OUs.
- Microsoft recommends using OUs rather than domains for structure and to simplify the implementation of policies and administration.
- The OU is the level at which administrative powers are commonly delegated, but delegation can be performed on individual objects or attributes as well.

Physical structure of AD

I) Site

- Sites in Active Directory represent the physical structure, or topology, of your network.
- You use Active Directory Sites and Services to define sites and site links.
- A site is a set of well-connected subnets.
- Sites differ from domains; sites represent the physical structure of your network, while domains represent the logical structure of your organization.

II) Domain controller

- A *domain controller* is a computer running Microsoft Windows Server 2012 that stores a replica of the domain directory.
- A **domain controller** (DC) is a server that handles all the security requests from other computers and servers within the Windows Server domain.
- Having more than one domain controller in a domain provides fault tolerance.
- If one domain controller is offline, another domain controller can provide all of the required functions, such as recording changes to the Active Directory service.
- Domain controllers manage all aspects of users.

Cont.

- Members only have to log into one domain to access all resources in the network.
- In a trusted relationship, one domain may gain access to other domains.
- In this case, members who log into the first domain will have access to the resources in the other domains.
- **Backup Domain Controller (BDC)** is a computer that has a copy of the user accounts database.
- Unlike the accounts database on the Primary Domain Controller (PDC), the BDC database is a read-only copy.

Cont.

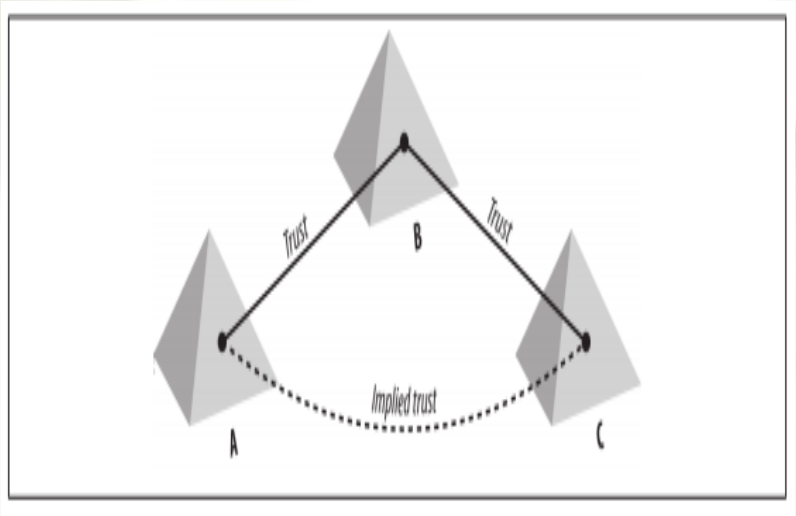
- When changes are made to the master accounts database on the PDC, the PDC pushes the updates down to the BDCs.
- Most domains will have at least one BDC, and often there are several BDCs in a domain.
- These additional domain controllers exist to provide fault tolerance.
- In such circumstances, an administrator promotes a BDC to be the new PDC.

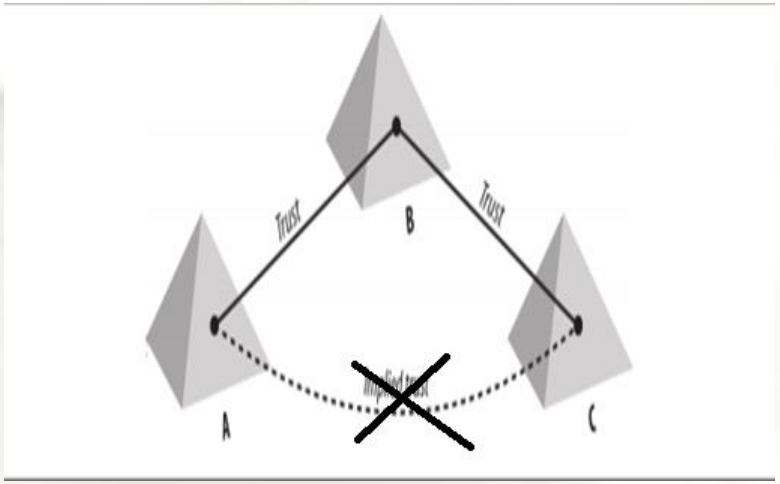
Trust relationship between two different domains

Trust: A relationship between different domains or forests that **allow sharing of resources between them.**

- Trust can be transitive or non-transitive.
- **Transitive Trust:** Trust which can be extended to other domains in the forests.
- **Non- Transitive Trust:** Trust which cannot be extended to other domains in the forests, it is only between the two domains of different forests.

Transitive & Non- Transitive Trust





Direction of Trust

- **One-way Trust:** Network A, trusts network B, and then network B can access network A only.
- **Two-way Trust:** Network A trusts network B, and vice versa, then both networks A and B can access each other.

Types of Trusts

There are several different types of trusts in Windows Server 2003/2008/2012. These are listed below:

- **Shortcut trust:** A shortcut trust is used to improve user logon times between two domains which are logically distant from each other in the Active Directory hierarchy.
- This trust is created manually and is transitive.
- It can also be either one-way or two-way.

External trust: An external trust is a trust created manually between domains in two separate forests or between a Windows Server 2008 domain and a domain running Windows NT 4.0 or earlier.

- External trusts are not transitive and can be either one-way or two-way.

Cont.

Realm trust: A realm trust is a trust created manually between a Windows Server 2012 domain and domain running on a non-Microsoft implementation e.g. **UNIX**.

- This trust can be either transitive, non-transitive, one-way or two-way.

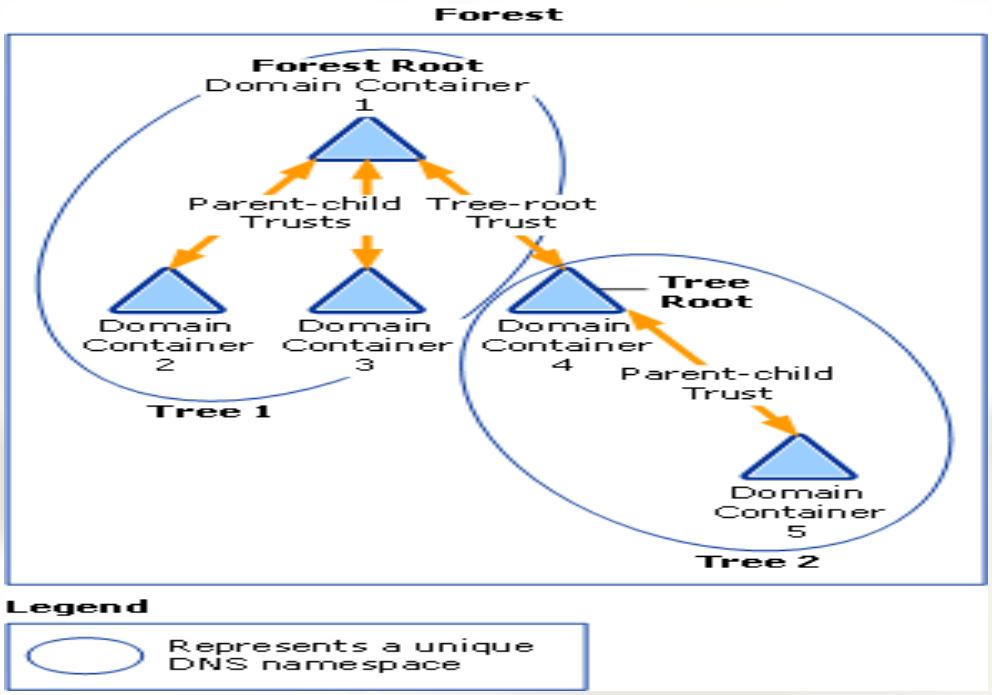
Forest trust: A forest trust is created manually between two Windows Server 2008 forests.

- The trust allows all domains in one forest to trust all domains in another forest, however a forest trust is not transitive across three or more forests.
- This trust can be either one-way or two way.

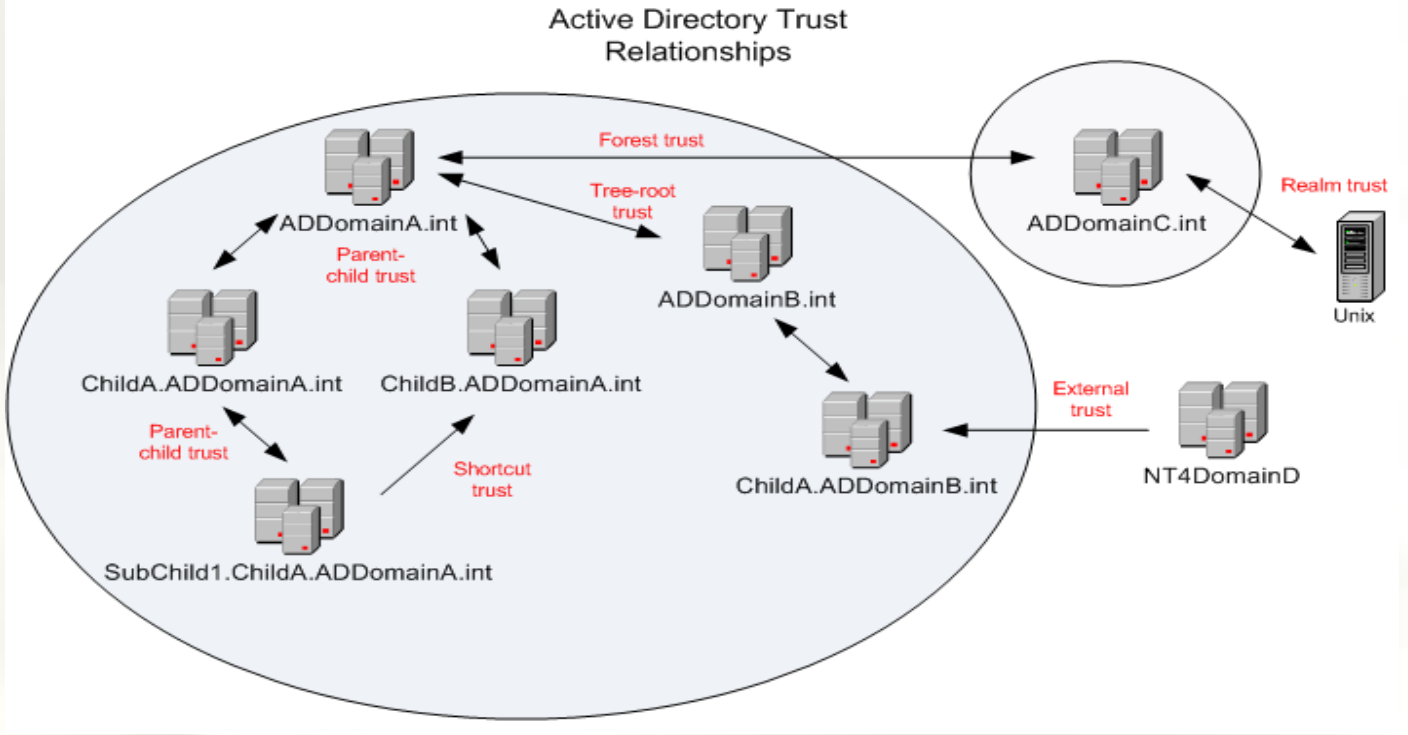
Cont.

- **Tree-root trust:** A tree-root trust is created automatically between a new tree and its root domain.
- This trust is transitive and two-way by default.
- **Parent-child trust:** A parent-child trust is created automatically between a child and its parent domain.
- This trust is transitive and two-way by default.

Cont..


Forest
Forest Root
Domain Container
1
Parent-child
Trusts
Tree-root
Trust
Tree
Root
Domain
Container
2
Domain
Container
3
Domain
Container
4
Parent-child
Trust
Tree 1
Domain
Container
5
Tree 2
Legend
Represents a unique
DNS namespace

Cont..


Active Directory Trust Relationships
Forest trust
Tree-root trust
Parent-child trust
ADDomainA.int
ADDomainB.int
ADDomainC.int
Realm trust
Unix
ChildA.ADDomainA.int
ChildB.ADDomainA.int
Parent-child trust
Shortcut trust
SubChild1.ChildA.ADDomainA.int
ChildA.ADDomainB.int
External trust
NT4DomainD

Dynamic Host Configuration Protocol (DHCP)

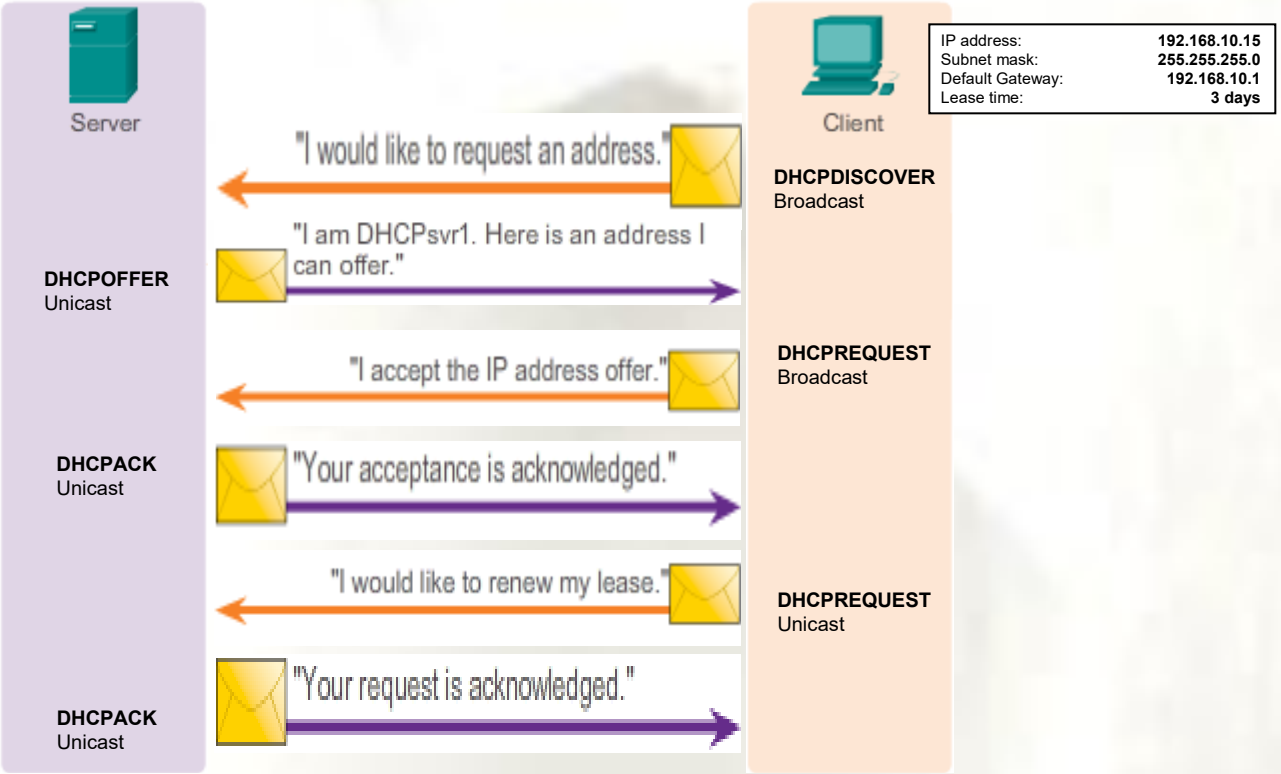
The Dynamic Host Configuration Protocol (DHCP) is a set of rules used by a communication device such as a computer, router or network adapter to allow the device to request and obtain an IP address from a server which has a list of addresses available for assignment.

- IP addresses and other parameters such as the **default gateway, subnet mask, and IP addresses of DNS servers** can be assigned by the DHCP server.
- The DHCP server ensures that all IP addresses are unique.
- IP address assignment occurs when the DHCP **client boots up or regains connectivity** to the network.

DHCP Process

- The DHCP client sends out a query requesting
- The DHCP server then replies to the client with its assigned IP address, subnet mask, DNS server and default gateway information.
- Lease IP addresses for a specific amount of time to DHCP clients, and then automatically renew the IP addresses when the client requests a renewal.

Cont …


Server
Client
IP address: 192.168.10.15
Subnet mask: 255.255.255.0
Default Gateway: 192.168.10.1
Lease time: 3 days
"I would like to request an address."
DHCPDISCOVER
Broadcast
"I am DHCPsvr1. Here is an address I can offer."
DHCPOFFER
Unicast
"I accept the IP address offer."
DHCPREQUEST
Broadcast
"Your acceptance is acknowledged."
DHCPACK
Unicast
"I would like to renew my lease."
DHCPREQUEST
Unicast
"Your request is acknowledged."
DHCPACK
Unicast

Benefits of DHCP

Reliable IP address configuration.

DHCP minimizes configuration errors caused by manual IP address configuration, such as address conflicts caused by the assignment of an IP address to more than one computer at the same time.

Reduced network administration.

- Centralized and automated TCP/IP configuration.
- The ability to define TCP/IP configurations from a central location.

DHCP Configuration settings

- **Scope:** a range of IP addresses on your network to which DHCP services are offered.
- **Exclusion range:** is a limited sequence of IP addresses within a scope, **excluded from DHCP service offerings.**
- **Address pool:** After you define a DHCP scope and apply exclusion ranges, the remaining addresses form the available *address pool* within the scope. And becomes eligible for dynamic assignment
- **Lease:** **is a length of time that a DHCP server specifies**, during which a client computer can use an assigned IP address.
- **Option types:** are other client configuration parameters a DHCP server can assign when serving leases to DHCP clients.

DNS Server Overview

Domain Name System (DNS) is a hierarchical distributed naming system for computers, services, or any resource connected to the Internet or a private network.

" When a user enters the DNS name of a computer in an application, DNS clients and servers work together to look up the name and provide other information that is associated with the computer, such as its IP address or services that it provides for the network. This process is called name resolution. "

Domain name syntax

- *A DNS name consists* of two or more parts separated *by periods, or "dots" (.).* The last (rightmost) part of the *name is called the top-level domain (TLD).*
- Other parts of the name are called subdomains of the TLD or another subdomain. The names of the TLDs are either *functional or geographical.* for example, the domain

– name *www.example.com* belongs to the top-level domain *com*.

Cont ...

A ***hostname*** is a domain name that has at least one IP address associated.

For example, the domain names www.example.com and example.com are also hostnames, whereas the com domain is not.

DNS consists of:

A) **Hierarchical name space**: name allocation decentralized to domains

host.sub-subdomain.. . ..subdomain.domain[.ROOT]

Examples of Fully Qualified Domain Names (FQDNs):

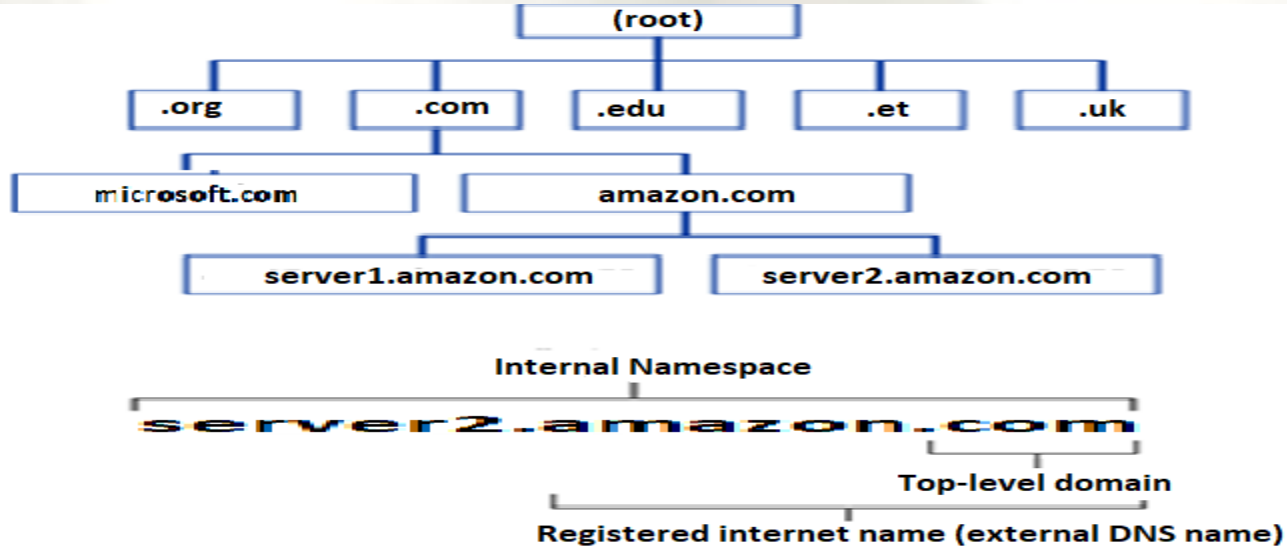
www.iot.uog.edu, www.coi.uog.edu

Cont ..

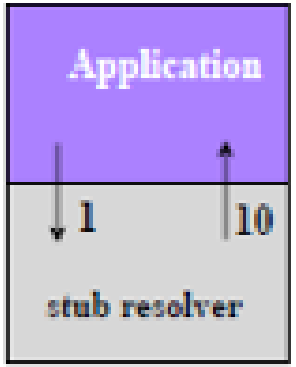
B) **Hierarchical name resolution infrastructure:**

- ✓ A distributed database storing resource records (RRs
- ✓ *Client-server query-reply*

Berkeley Internet Name Domain (BIND): the most common implementation of the DNS name resolution architecture



DNS Name Resolution



Client:

- ✓ Has stub resolver linked in
- ✓ Consults /etc/resolve.conf to find local name server
- ✓ Form FQDN

Local name server:

- ❑ Receive a DNS query from host
- ❑ Acts as a proxy, forwards query into hierarchy
- ❑ Parses FQDN from right to left
- ❑ Consults its caches to find root name server
- ❑ Caches resolved name (like TLD name server and others) to reduce latency in DNS translation.

Types of Query for DNS name resolution

Recursive query:

- Local name server must resolve the name (or return "not found")
- Put burden of name resolution on contacted name server

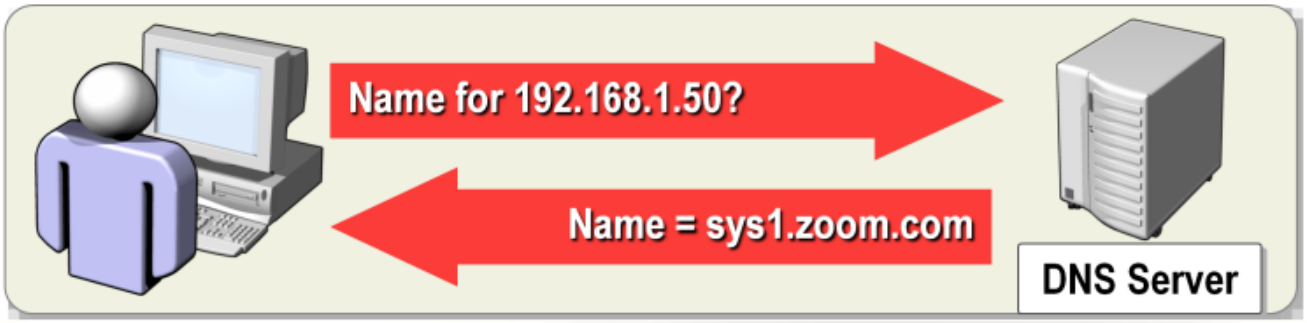
Iterative query:

- Contacted server replies with the name of server address of sub-domain ("i dont know this name, but ask this other name server")
- Requesting name server visits each name server referred to

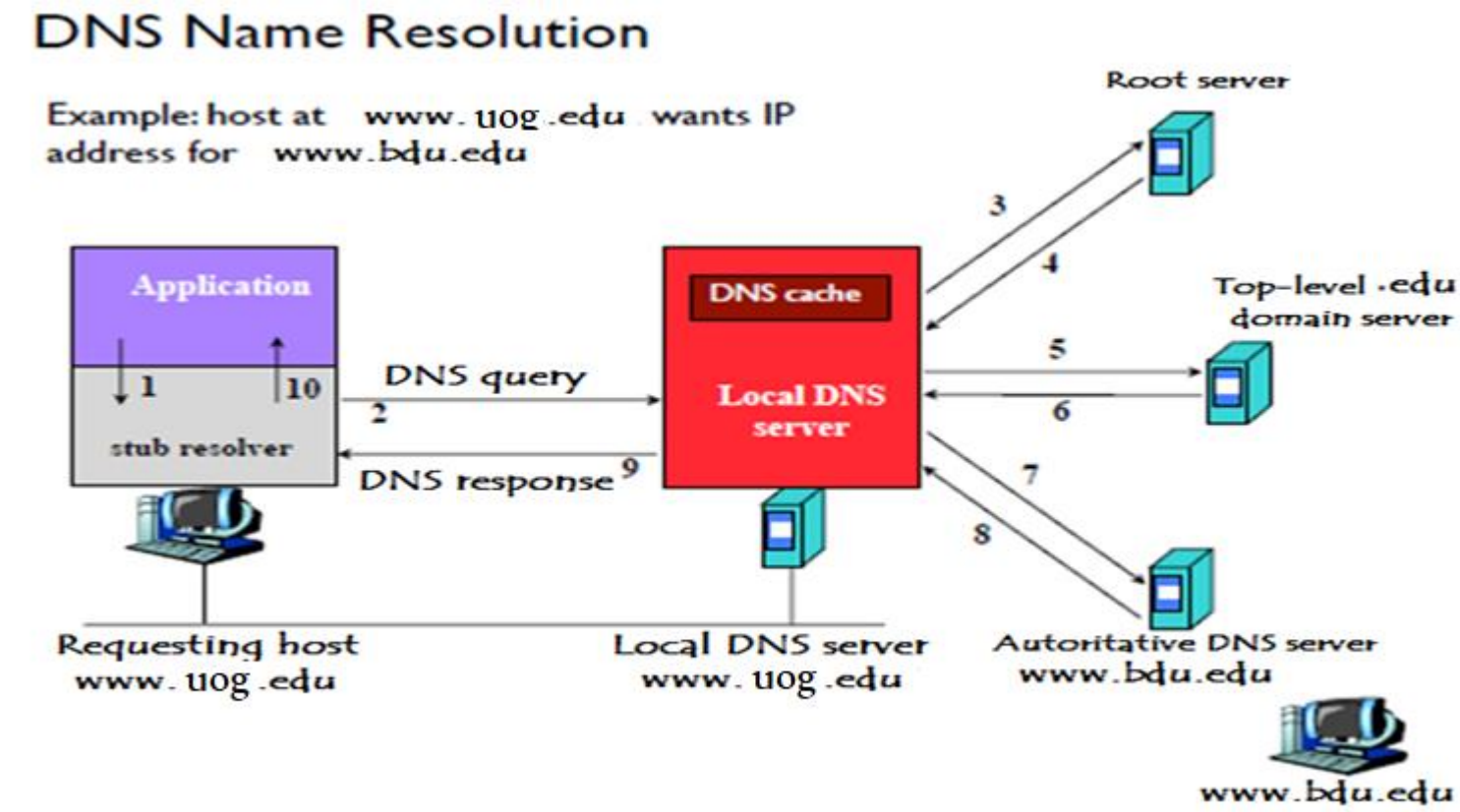
DNS Lookup Zone types

- **Forward Lookup**
  - Requests Name-to-IP Address resolution

- **Reverse Lookup**
  - Requests IP Address-to-Name resolution



Example


DNS Name Resolution
Example: host at www.uog.edu wants IP address for www.bdu.edu
Root server
Application
1
10
stub resolver
DNS query
2
DNS response
9
DNS cache
Local DNS server
3
4
Top-level .edu domain server
5
6
7
8
Requesting host www.uog.edu
Local DNS server www.uog.edu
Autoritative DNS server www.bdu.edu
www.bdu.edu

Lightweight Directory Access Protocol (LDAP)

- LDAP is an application protocol used to access and maintain directory services over a network.
- LDAP is **an open and cross platform protocol used for directory services authentication**.
- LDAP provides the communication language that applications use to communicate with other directory services servers
- Deployed on the server that is not a domain controller.
- LDAP is based on the client/server model of distributed computing

Cont.

LDAP defines operations for accessing and modifying directory entries such as:

- Searching for entries meeting user-specified criteria
- Adding an entry
- Deleting an entry
- Modifying an entry
- Modifying the distinguished name or relative distinguished name of an entry (move)
- Comparing an entry

Questions ?

User administration concepts and mechanism

Chapter 3

Objectives

- To describe user accounts, user profile, & backup
- To define roaming, mandatory and local profile
- To list and describe different types of user account
- To list and describe different types of user profile
- To list and describe different types of backup
- To describe windows registry

Introduction

- Mostly users have read only access to all components of the product. But they will not have the privileged to access, configure, or edit different documents.
- But system administrators are allowed to perform all administrative activities.
- The primary difference between a **standard user** and an **administrator** is the level of access that the user has over core, protected areas of the computer.

Cont. ..

- Administrators can change the system state, turn off the firewall, configure security policies, install a service or a driver that affects every user on the computer, and install software for the entire computer.
- Standard users cannot perform these tasks, and they can only install per-user software.
- Active directory authentication module is implemented in application manager in order to ease the user administration.

Cont..

- **User Account Control (UAC)** is a security component that enables users to perform common tasks as **non-administrators (called standard users),** and **as administrators** without having to switch users, log off, or use Run As.
- **User accounts** that are members of the local Administrators group run most applications as a **standard user.**
- By separating user and administrator functions, UAC helps users move toward using standard user rights by default.

User Accounts

Account determines 3 factors:

- when a user may log on
- where within the domain/workgroup
- what privilege level a user is assigned

Types of user account

1. Local Accounts

- **Local account** is an account setup on the machine.
- It is maintained on the local system, not distributed to other systems
- **Local user account** authenticates the user for local machine access only; access to resources on other computers is not supported
- In Windows, a local user is one whose username and encrypted password are stored on the computer itself.
- When you log in as a local user, the computer checks its own list of users and its own password file to see if you are allowed to log into the computer.

II. Domain Accounts

- **Domain account** is when you login using a domain account on a network.
- You would need a domain controller to be able to setup accounts to login.
- You would then join the computer to the domain and use a login from the central database.
- Permit access throughout a domain and provide centralized user administration through AD

User Profiles

A user profile is a collection of a user's personal files and settings that define his or her working environment. (like Documents, Desktop, Music, Pictures, & Favorites)

Types of User profiles:

- Local profile
- Roaming profile
- Mandatory Profiles

Local profile

- Local profiles are created from a default profile when the user first logs on to a specific machine.
- Changes on one local profile will not migrate to another local profile on another machine
- Any changes that are made to a local profile are *kept only on that machine*.

Mandatory Profiles

A mandatory user profile is a special type of pre-configured roaming user profile that **administrators can use to specify settings for users.**

User profiles become mandatory profiles when the administrator renames the NTuser.dat file on the server (location: C:\Windows\System32\config\systemprofile) to NTuser.man.

There are two types of mandatory profiles: **normal mandatory** profiles and **super-mandatory profiles.**

Cont ...

- User profiles become super-mandatory when the folder name of the profile path ends in .man; E.g \\server\share\mandatoryprofile.man\.
- users who have **super-mandatory** profiles cannot log on when the server that stores the mandatory profile is unavailable, but users with **normal mandatory** profiles can log on with the locally cached copy of the mandatory profile.

Major characteristics of mandatory profile

- Used when you don't want users to be able to change their profile, or only have the ability to make temporary changes
- Mandatory profiles are fixed profile in which the user changes will not be saved
- Only administrators can make changes to mandatory profiles.

Roaming user profiles

Roaming user profiles are the profiles of ***domain users*** which are saved on a file server located anywhere in the ***domain environment.*** With the help of roaming profiles domain users are able to get *their desktop environment on every computer* which is client of a domain controller.

Why we need to have roaming profile?

- To avoid different desktop ambiance for the user
- Automatic resource availability.
- Simplified computer replacement and backup

Backup and Restore

- If the ***data is destroyed or corrupted***, the only hope you have of ***retrieving*** the data is from a ***properly configured backup***. A ***backup*** is simply a ***copy of the data you have*** on ***network share or a tape device.***
- Backups have two distinct purposes. The primary purpose is **to recover data after its loss**. The secondary purpose of backups is **to recover data from an earlier time**, according to a user defined.

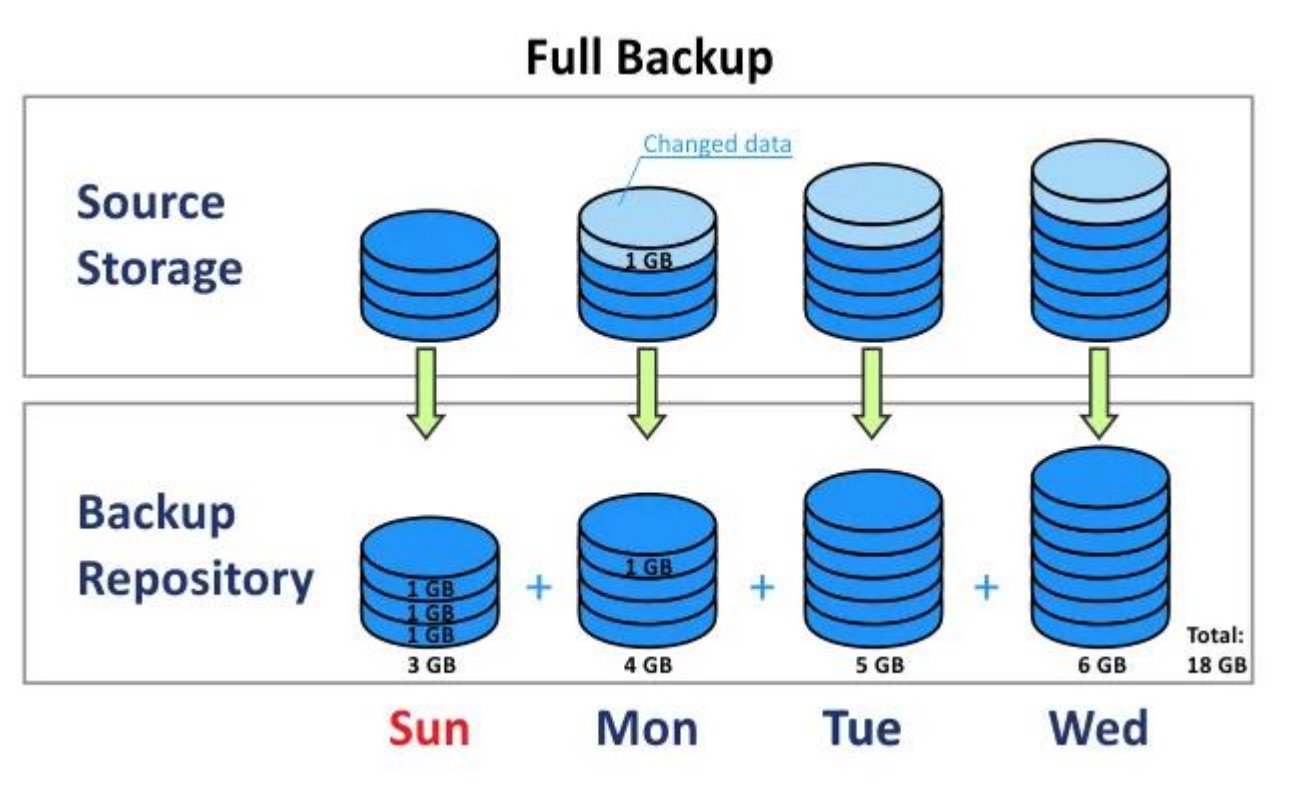
Types of backup

There are **several types of backups** that can be performed: ***normal, differential, and incremental***.

- ***A normal backup***, also known as a ***full backup***, will copy all the designated files.

- This type of backup takes the ***longest*** to complete, but is the ***quickest to restore***. Since there is usually ***only one media item that contains the full backup***, only one is needed to restore and as such is the ***quickest to restore***

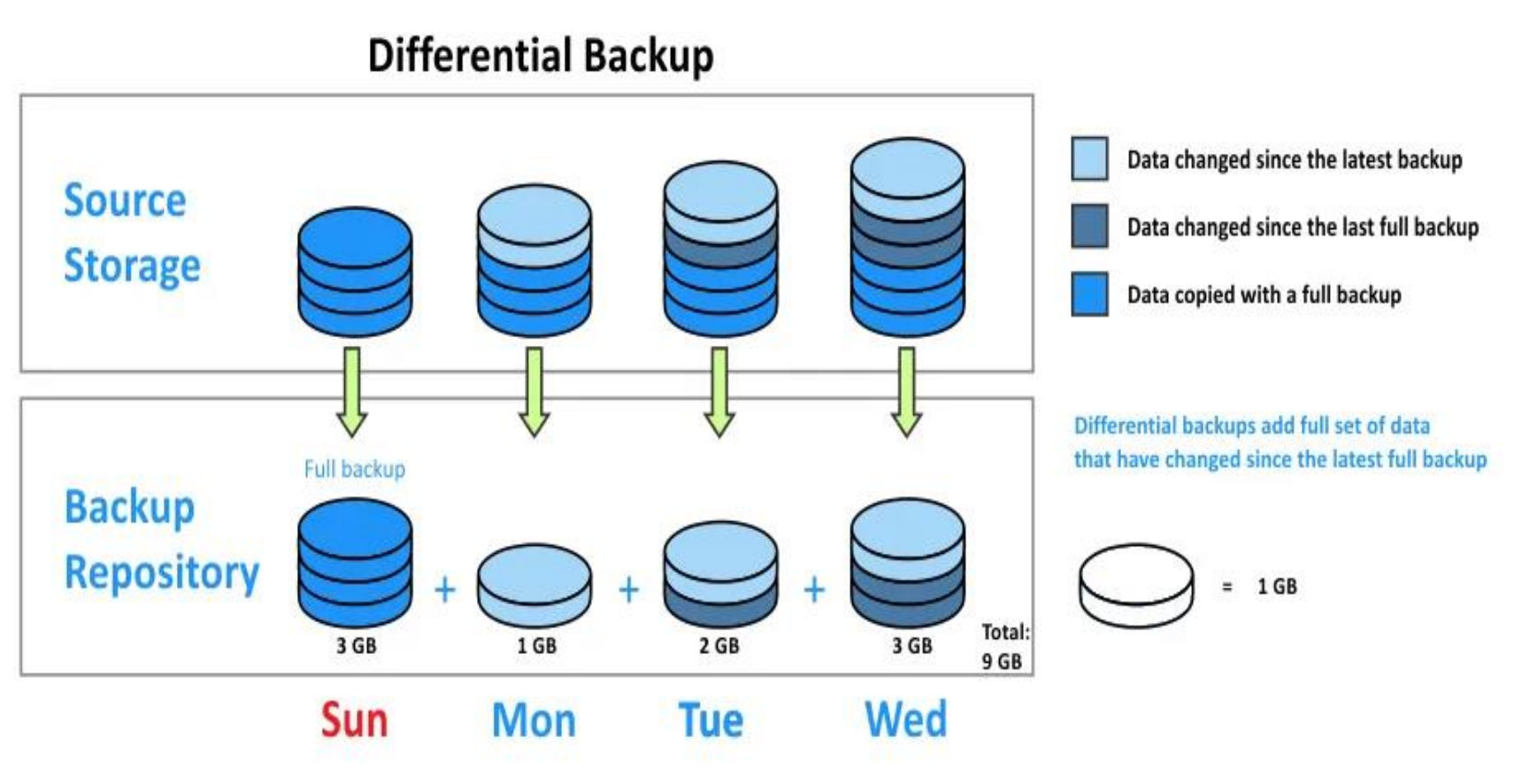
Cont. ..


Full Backup
Changed data
Source Storage
1 GB
Backup Repository
1 GB
1 GB
1 GB
+
1 GB
+
+
3 GB
4 GB
5 GB
6 GB
Total: 18 GB
Sun
Mon
Tue
Wed

Cont. ..

- A ***differential backup*** copies all the files that have ***changed since the last full backup***.
- In terms of backup/restore speed, the differential backup type has some advantages:
  - Backups are completed faster than full ones.
  - Consumes less storage space than a full backup.
  - Data recovery is faster than using an incremental backup with many increments.
- The disadvantages of differential backup are:
  - Recovery is slower than when using a full backup since there will be ***two media items to restore***: ***the full backup media and the differential backup media***.
  - Not the most optimal backup type in terms of storage space consumption.

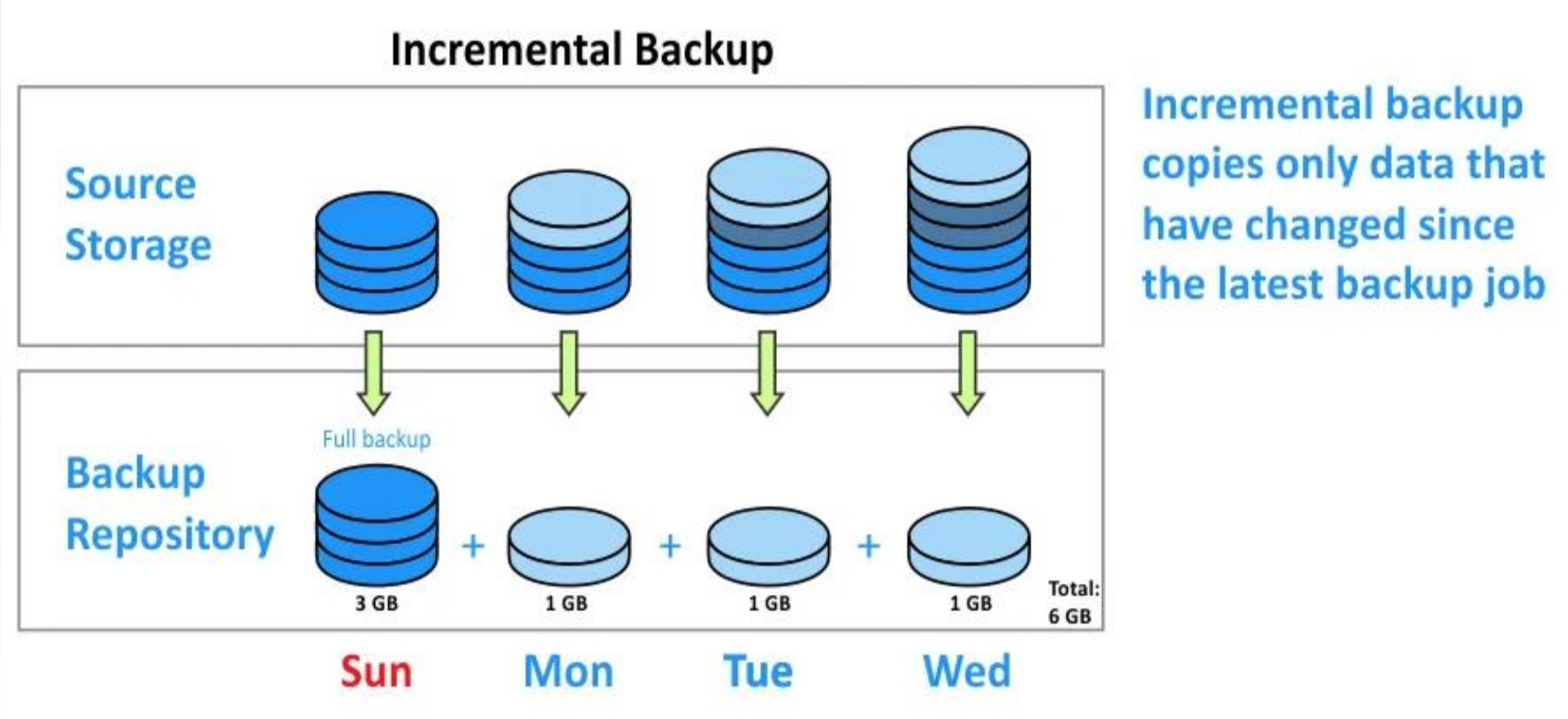
Cont. ..


Differential Backup
Source Storage
Backup Repository
Full backup
3 GB
1 GB
2 GB
3 GB
Total: 9 GB
Sun
Mon
Tue
Wed
Data changed since the latest backup
Data changed since the last full backup
Data copied with a full backup
Differential backups add full set of data that have changed since the latest full backup
= 1 GB

Cont. ..

- ***An incremental backup*** backs up the data since the ***last backup***, whether full or incremental.
- This means that if you did an ***incremental backup each day***, you would ***only back up the files that changed on that day***.
- As a result, the ***backup times are usually shorter***. However, restoring can ***take much longer***.
- Depending on ***how many incremental backups were done since the last full backup***, the ***restore process will take longer and be more tedious***.

Cont. ..


Incremental Backup
Source Storage
Backup Repository
Full backup
3 GB
1 GB
1 GB
1 GB
Total: 6 GB
Sun
Mon
Tue
Wed
Incremental backup copies only data that have changed since the latest backup job

Cont. ..

Comparison Table of 3 Main Backup Types

	Full	Differential	Incremental
Storage space use	High	Medium to high	Low
Backup speed	Slow	Medium	Fast
Recovery speed	Very fast	Fast	Slow
Recovery requirements	The most recent backup or relevant backup for the required point in time	One full and one differential backup	Full backup and all subsequent incremental backups
Ease of use	Easiest	Medium	Medium

Windows registry

The Windows registry is a central repository of information about all aspects of the computer in particular, its hardware, operating system, applications and users.

Regedit is the Windows registry editor, a graphical tool that lets you view and monitor the Windows operating system's registry and edit if necessary.

Because the Windows registry controls important configuration settings on your computer, you shouldn't try to access or edit it without knowing exactly what you're doing.

The registry contains two basic elements: keys and values.

How does the Windows registry work

Registry keys are containers that act like folders, with values or subkeys contained within them. **Registry values** are similar to files (not containers).

Not all applications use the registry. Some apps use XML files or local executable files to store their settings.

The main branches of the registry are called **hives**. And most PCs have five of them. All the folders in the registry are called *keys* except for these five hives. Inside these hives are more folders called keys. Keys contain values, which are the settings themselves. The hives in the Windows registry have the following names:

Cont.

HKEY_CLASSES_ROOT(HKCR) — keeps track of default file associations. This key contains information about registered applications

HKEY_CURRENT_USER (HKCU) — contains settings specific to your username in Windows.

HKEY_LOCAL_MACHINE (HKLM) — contains passwords, boot files, software installation files, and security settings. This is the most critical hive.

HKEY_USERS (HKU) — like the CURRENT_USER hive, except it's for when more than one user is logged onto the server or computer.

HKEY_CURRENT_CONFIG (HKCC) — This key contains information gathered at runtime; information stored in this key is not permanently stored on disk, but rather regenerated at boot time.

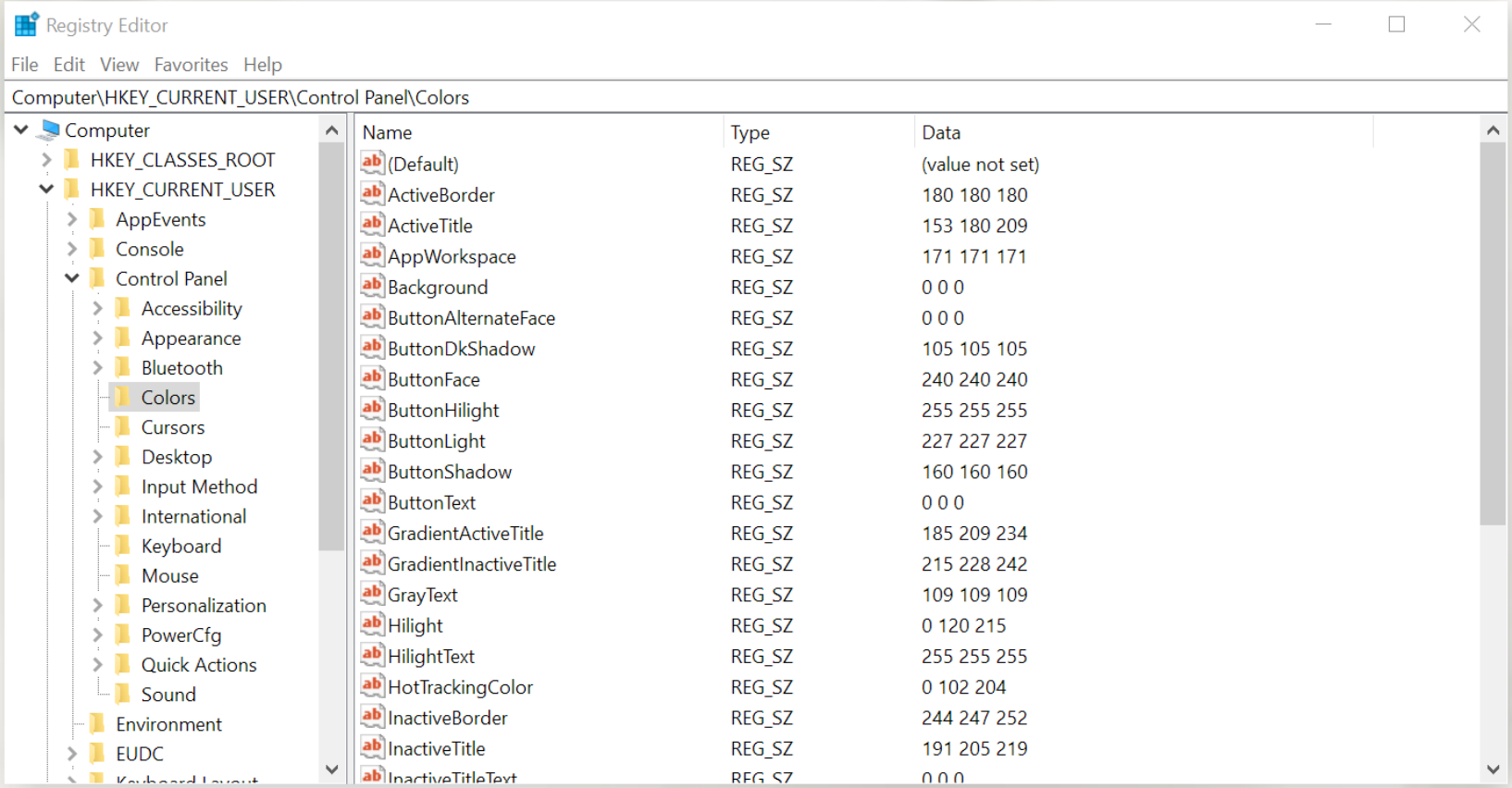
When to use the Windows registry

You may need to use the Windows registry to fix performance issues, like if your computer keeps crashing. Or, you might want to edit the registry to change parts of your user experience, like your desktop settings. Sometimes, items in your registry might be broken, in which case you should use a registry cleaner from a trusted software provider.

Before making any changes, always create a backup of the registry a backup doesn't take up much space and will save you in case something goes wrong.

Cont'd…

Here is the registry editor in Windows 10, showing individual, detailed color settings:



Chapter 4

Objectives

- To explain Resources Monitoring
- To describe Network Management
- To describe TCP/IP Troubleshooting
- To describe IP addressing
- To explain File, Mail, & Proxy Servers
- To describe Remote Administration
- To understand Network Printing
- To describe SCSI & RAID

Resource monitoring

Resource monitoring is a system application that used to view information about the use of **hardware** (CPU, memory, disk, and network) and **software** resource in real time.

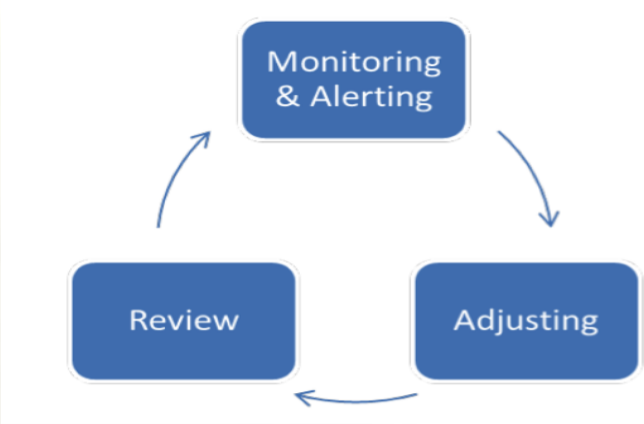
System Performance Monitoring

- Either the system is running too slowly, or programs (and sometimes even the entire system) fail to run at all.
- System performance monitoring is normally done in response to a performance problem.
- A system performance monitor (SPM) is a type of application that identifies, collects, monitors and reports on the overall operational health of a computer system.

Cont..

Some of the performance metrics/data collected by an SPM include the use of the CPU, memory, hard disk and network.

3 Steps to Successful Performance Management:



- Monitoring to identify the nature and scope of the resource shortages that are causing the performance problems
- The data produced from monitoring is analyzed and a course of action (normally performance tuning and/or the procurement of additional hardware) is taken to resolve the problem
- Monitoring to ensure that the performance problem has been resolved

Monitoring CPU Power

- A CPU is generally a single microprocessor made from a semiconducting material, usually **silicon**, with millions of electrical components on its surface.
- Standard CPUs contain processing units that interpret and implement software instructions.
- Perform calculations and comparisons, make logical decisions
- If CPU utilization stays below 100%, no matter what the system is doing, there is additional processing power available for more work.

Monitoring Memory

- A computer stores a number into a specific location in memory and later fetches the value.
- In any case, the following statistics are useful for monitoring storage:
  - **Free space** is probably the one resource all system administrators watch closely; It would be a rare administrator that never checks a free space (or has some automated way of doing so).
  - **File System-Related Statistics** These statistics (such as number of files/directories, average file size, etc.) provide additional detail over a single free space percentage.

Monitoring Bandwidth

- Bandwidth describes the maximum data transfer rate of a network or Internet connection.
- While bandwidth is used to describe network speeds, **it does not measure how fast bits of data move from one location to another.**
- Since data packets travel over electronic or fiber-optic cables, the speed of each bit transferred is negligible.
- Instead, **bandwidth measures how much data can flow through a specific connection at one time.**
- **In networking,** bandwidth represents the overall capacity of the connection.

Network Management

Network Management: refers to the broad subject of managing computer networks. Network management covers a wide area, including

- **Security:** Ensuring that the network is protected from unauthorized users.
- **Performance:** Eliminating bottlenecks in the network.
- **Reliability:** Making sure the network is available to users and responding to hardware and software malfunctions‘

It generally includes the following tasks:

✓Fault detection for networks, gateways, and critical servers.

Cont..

- ✓Schemes for notifying an administrator of problems.
- ✓General network monitoring, to balance load and plan expansion.
- ✓Documentation and visualization of the network.
- ✓Administration of network devices from a central site.

Network Monitoring

Describes the use of a system that **constantly monitors** a computer network for **slow or failing components** and that notifies the network administrator in case of outages via email or other alarms.

TCP/IP Troubleshooting

Several tools are available for debugging a network at the TCP/IP layer:

PING: check to see if a host is alive or not

It sends an ICMP ECHO_REQUEST packet to a target host and waits to see if the host answers back.

NETSTAT : get network statistics

Collects information about the state of our computer's network, including **protocol**, **local address**, **routing information**, and **state etc**.

Cont..

Ipconfig: Displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings.

Ifconfig: Used from the command line for Unix-like operating system either to assign an address to a network interface or to configure or display the current network interface configuration information.

TRACEROUTE and TRACERT:

Exposes the sequence of gateways/path details through which an IP packet travels to reach its destination. **E.g. tracert** google.com

IP Addressing

- unique and universal for global communication (no 2 hosts have the same address, but a host can have 2 IP addresses)
- with 32-bits there are 2^{32} (4,294,267,296) hosts; now not sufficient due to a number of reasons (mainly wastage)

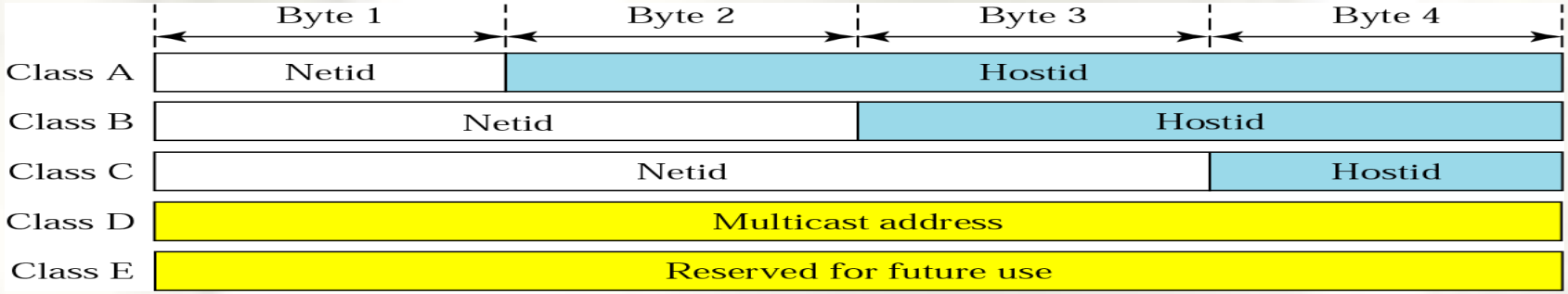
Classful Addressing

- address space is divided into 5 classes: A, B, C, D, and E
- Addresses in classes A, B, and C are for unicast communication;

Cont ...

- Addresses in class D are for multicast communication (only as a destination); if a host belongs to a group or groups, it can have one or more multicast addresses
- Addresses in class E are reserved; the idea was to use them for special purposes

- **Netid and Hostid**



Address Depletion

- The flaws in classful addressing scheme combined with the fast growth of the Internet led to the near depletion of the available addresses.
- We have run out of class A and B addresses, and a class C block is too small for most mid size organizations.
- Some solution that has alleviated the problem is the idea of classless addressing, Subnetting, & private address

CIDR (Classless Inter-Domain Routing)

Classless addressing an addressing mechanism in which the IP address space is not divided into classes. To overcome address depletion and give more organizations access to the Internet.

CIDR identifies the number of binary bits set to a **1** (or on) in a subnet mask, preceded by a slash.

E.g, a subnet mask of 255.255.255.240 would be represented as follows in binary:

11111111.11111111.11111111.11110000

The first 28 bits of the above subnet mask are set to 1. The CIDR notation for this subnet mask would be /28.

Subnetting

- **Subnetting** the act of dividing a network into smaller logical sub network. stealing bits from hosts creates more networks but fewer hosts per network.
- To calculate the number of subnet using the following formula: $x=2^n$('**n**' identifies the number of bits to borrowed from the host portion of the subnet mask).
- The formula for calculating **usable hosts per subnet** is $x=2^n-2$ ('**n**' identifies the number of unborrowed bits) Because it is never possible to assign a host an address with all 0 or all 1 bits in the host portion of the address.

A private address

- **A private address** is intended for internal use within a home or organization, and can be freely used by anyone. However, private addresses can never be routed on the Internet.
- Three private address ranges class:

Range			*Total*
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

Example

- **A company would like to break its Class B network IP address 172.16.0.0 into 60 different subnets. Find ranges of IP addresses for each subnet and new mask.**
  - Class B network has 16 host bits → Class B subnet mask = 255.255.0.0 = 11111111.11111111.00000000.00000000
  - 60 = 00111100 → we need at least 6 additional network bits
  - New Mask 11111111.11111111.11111(1)00.00000000=255.255.252.0 and bit with parenthesis is the increment bit
  - Start with the given network IP address and add the increment to the subnetted octet:

    172.16.0.0

    172.16.4.0

    172.16.8.0 … etc.

Cont ..

- Now add each end range, which is the last possible IP address before the next range :

Subnet 1	172.16.0.0	–	172.16.3.255
Subnet 2	172.16.4.0	–	172.16.7.255
Subnet 3	172.16.8.0	–	172.16.11.255
Subnet 4	172.16.12.0	–	172.16.15.255
	...		
	...		
Subnet 60	172.16.236.0	–	172.16.239.255

- Assign these ranges to the new networks, but the first and last address from each range (network / broadcast IP) are unusable.

An organization would like to break its class C private IP address 192.168.5.0 into as many subnets as possible provided that they can get at least 30 clients per subnet. Find ranges of IP addresses for each subnet and new mask.

HOME WORK ....

A company would like to break its Class C private IP address 192.168.5.0 into as many subnets as possible provided that they can get at least 31 clients per subnet. Find ranges of IP addresses for each subnet and new mask.

File, Mail, & Proxy Servers

File server

- In the client/server model, **a file server is a computer responsible for the central storage and management of data files** so that other computers on the same network can access the files.
- A file server provides a central location on your network where you can store and share files with users across your network.

Mail server

- **Email server** is software or program that can send or receive email from or to other email server.
- User names and host names in email address creation are separated by **"@".**

Types of mail services

1) Outgoing mail server
2) Incoming mail server

- **Outgoing mail server** is used to send mails from the sender to the receiver with the helps of protocol like SMTP.
- **Incoming email server** is used to receive mail messages from the sender. E.g POP3, IMAP

Cont.

IMAP (Internet Mail Access Protocol) and POP3 (Post Office Protocol)

- Both used to receive email from the sender.
- POP3 is used for storing sent and received messages on PCs local hard drives.
- IMAP is always used to store copies of messages on servers.
- Main differences between IMAP and POP3:
  - POP3 protocol assumes there is only one client connected to the mailbox
  - IMAP protocol allows simultaneous access by multiple clients, so it is suitable if the mailbox is used by multiple users.

Proxy Server

A **proxy server** is a system or router that provides a gateway between users and the internet. & provide increased performance and security.

It is a server, referred to as an “intermediary” because it goes between end-users and the web pages they visit online.

A **proxy server**, also known as a "**proxy**" or "application-level gateway", is a computer that acts as a gateway between a local network.

Proxies provide a valuable layer of security for your computer. They can be set up as web filters or firewalls, protecting your computer from internet threats like malware. Screening all incoming and outgoing traffic.

Advantages of Proxy server

An advantage of a proxy server is that its cache can serve all users. If one or more Internet sites are frequently requested.

these are likely to be in the proxy's cache, which will improve user response time.

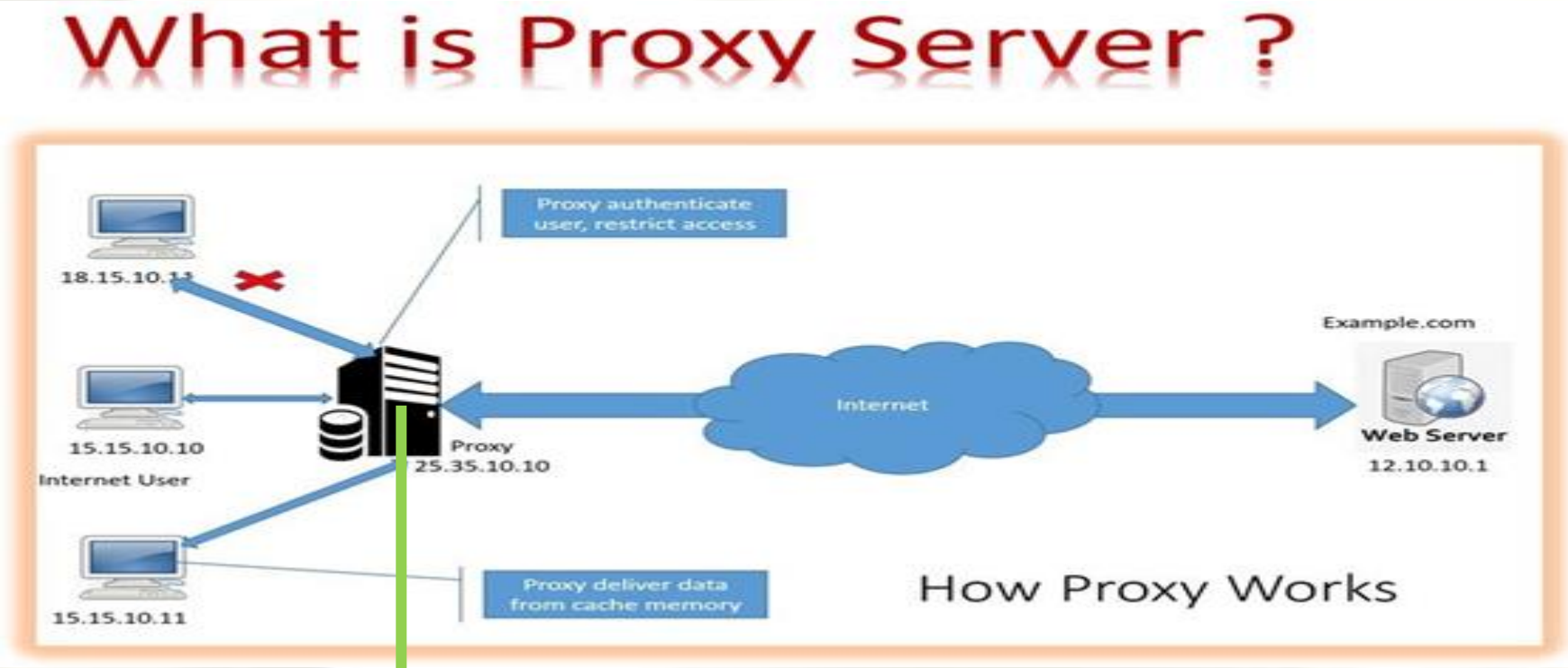
- **Privacy**: it hides your IP address, without using proxy server, your public IP address is visible.
- **Activity logging**: a proxy server will keep a record of what website they are visiting and how long they are were on those websites. which can be helpful for troubleshooting.
- **Speed, bandwidth saves**

Here's a simple example of how proxy servers work

When a proxy server receives a request for an Internet resource (such as a Web page), it looks in its local cache of previously pages.

- If it finds the page, it returns it to the user without needing to forward the request to the Internet.
- If the page is not in the cache, the proxy server, acting as a client on behalf of the user, uses one of its own IP addresses to request the page from the server out on the Internet.
- When the page is returned, the proxy server relates it to the original request and forwards it on to the user.

how proxy servers work



IP Address	Activity website	Logging time

Remote Administration Services

- Remote administration is any method of controlling a computer from a remote location.
- Remote control software's has the ability to login and access computers remotely.
- ***Remote desktop*** refers to a software or operating system feature that allows a personal computer's desktop environment to be run remotely. Other E.g GoToMypc, PcAnywhere and RemotelyAnywhere.

Common tasks of remote administration.

1) **General**: controlling one's own computer from a remote location.

2) **Shutdown**: is the process for **shutting** another computer over a network.

3) **Accessing peripherals**: it is possible to access network resources like **printer** remotely.

4) **Modifying**: remote administrators can modifying system services, installing software on another machine, and resetting passwords.

5) **Viewing**: like remotely assisting others or supervising computers.

Network Printing

- Network printing is the collection of software components in Windows Server 2012 that **provide network printing services for client computers.**
- In some small offices, it might be practical to have a printer attached to every computer, but in an organization consisting of more than a few computers, sharing printing resources is desirable.
- Network administrators **must manage** network printing, printer availability, and printer security
- A print device can be accessed through a print server
  - Using a **local print device**
  - Using **network print devices**

Cont..

Local Print Devices

- It is essential to install the necessary **drivers** to ensure proper working of the print device.
- Local print devices provide the most **convenient/suitable** way of printing from a workstation computer.

Shared Print Devices

- Sharing a locally attached printer.
- Sharing print devices directly connected to the network.

SCSI (Small Computer System Interface)

- is a set of American National Standards Institute (ANSI) standard electronic interfaces that allow personal computers (PCs) to communicate with peripheral hardware such as disk drives, tape drives, CD-ROM drives, printers and scanners faster and more flexibly than previous parallel data transfer interfaces.
- The SCSI standard defines a bus along which multiple devices may be connected.
- A SCSI bus is a parallel bus, meaning that there is a single set of parallel wires that go from device to device.

Cont.

- Because these wires are shared by all devices, it is necessary to have a way of uniquely identifying and communicating with an individual device.
- This is done by assigning each device on a SCSI bus a unique numeric address or *SCSI ID*

Serial-attached SCSI (SAS)

- SAS products are compatible with devices that employ earlier SCSI technologies.
- SAS has become a popular alternative to parallel SCSI in enterprise environments. Both serial and parallel SCSI are based on the SCSI command set.

Cont.

SAS offers the following distinct advantages over parallel SCSI:

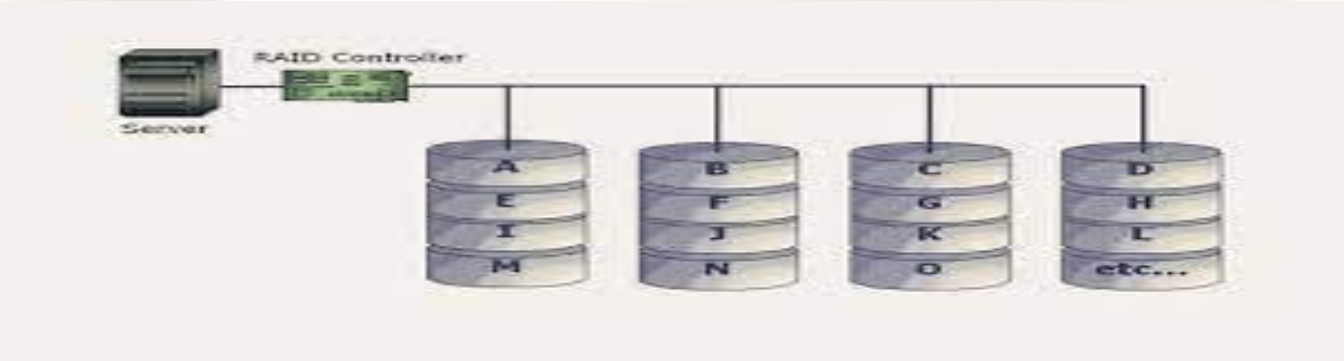
- It supports up to 65,535 devices (through the use of expanders). The latest parallel SCSI standards allow for only 16 devices.
- it eliminates issues with termination and clock skew.
- It is a point-to-point technology. This means SAS is not subject to the resource contention issues that are so common with parallel SCSI.

RAID-Based Storage

RAID(Redundant Array of Independent Disks) is a technology that is used to increase the performance and/or reliability of data storage, which is older and less used.

RAID is a way of storing the same data in different places on multiple hard disks or solid-state drives (SSDs) to protect data in the case of a drive failure.

RAID Controller: is a device that used to manage hard disk drives in a storage array so they work as a logical unit.



Questions ?

Introduction Security & Linux System Administration

CHAPTER FIVE

Objectives

- To describe **System Security**
- To describe **Security Mechanism**
- To define **Password Security**
- To list types of **Access Control and Monitoring**
- To define **Network Security Firewalls**
- To describe **Permissions and Type of Permissions**
- To describe **System Audits**
- To define **Linux Operating System**
- To describe LINUX **File System, Directories, File manipulation, & File Permissions**
- To describe **LINUX Process management, Users, & Groups managements**
- To demonstrate **File sharing on Linux using samba**

What is Security

- In general, **security** is "the **quality or state** of being **secure**--to be **free from any danger.**"
- **Thus, Security** is an important aspect of **operating system** design b/c it **safeguards against** access to resources by **unauthorized** users.
- The **security mechanism** can be **broken** in **two** steps: **authentication** and **authorization.**

Security Mechanism

- **Authentication:** Authentication a **mechanism** of **identification an individual**, is based on **what you know, what you have or who you are.**
- **Authorization**: a **mechanism of identification an individual privilege of access,** which is allowed to after authenticated the individuals. It asks, "**What are you allowed to do?"**

Password Security

- **Password Security** is the **first line** of **defense against** intruders.
- **Once** a **malicious user has gained access to an account**, it is very much **easier to exploit** other **weaknesses in security**.

Common Guidelines supported by Followers of Software System Security Password

- Use a minimum password **length** of 12 to 15 characters **if permitted**.
- Include **lowercase, uppercase alphabetic characters, numbers and symbols if permitted.**
- Generate **passwords randomly** where feasible.
- **Avoid** using the same password **twice**.
- **Avoid** character **repetition,** keyboard patterns, dictionary words, letter or number sequences, biographical information usernames.
- **Avoid using** information that might become **publicly associated** with the user or the account.
- **Avoid using** information that the **user's colleagues** might know to be associated with the user.
- **Do not use passwords** which consist only any **simple combination** of the aforementioned weak components.

Access Control and Monitoring

- ***Access control*** is a more general way of talking about **controlling/limiting access to a resource.**
- Access can be **granted** or **denied** based on a wide **variety of criteria**, such as:
  - The **network address** of the **client**,
  - The **identity** of the person who **wants access**, or
  - The **browser** which the **visitor** is using.
- If the **list says** that the **remote host** is **permitted** to use the **requested service**, the **access** is **granted.**
- If the **list says** that the **remote host** is **not permitted** to **access the service**, **access** is **denied.**

Access Control List (ACL)

- **An access control list (ACL)** is a **list** that **tells** a to the **operating system** which **access rights each user** has to a **particular system object,** such as a **file directory** or **individual file**.
- The **most common rights** include the **ability to read a file** (or all the files in a directory), **to write to the file**(s), and **to execute the file** (if it is executable)

Permissions

- **Permission** is the type of **access granted** to a **user, group,** or **computer** to **access resources.**
- **Permission** can be applied to **resources** such **files, folders** and **printers** like **privilege** to **read a file, delete a file,** or **to create a new file in folder.**

Types of permission

- Security level permission
- Share level permissions

Cont. ..

- **Security Level Permission**
  - Security or permission can set on **drives, folders** and **files.**
  - By default, security permissions will be inherited from its **parent drive** or **folder.**
  - Creators of **files** and **folders** are **owners.**

  Security permissions include: Full control, read and **execute, modify, write, read,** and list **folder** contents

- **Share level permission**
  - Can be set on **drives** and **shared folders** but **not files.**
  - **Shared folders** can be accessed from networks.
  - Share **permissions** include, **reader, owners** and **contributors.**

Network Security Firewalls

- **Firewall** is a **network security system** that **monitors** and **controls incoming** and **outgoing network traffic** based on **predetermined security rules.**
- A **firewall** typically **establishes** a **barrier** between a **trusted internal network** and **untrusted** outside network, such as the **Internet.**
- **Firewalls** are often categorized as either **network firewalls** or **host-based firewalls.**

System Audits

- **System Audits: -** It is a **review** on the **level** of **information security** in an **organization.**
- The **auditor** should **perform** the following work **before conducting** the **review:**
  - Meet with **IT management** to **determine** possible **areas** of **concern**
  - Review the **current** **IT organization chart**
  - Review **job** **descriptions** of **data center** employees
  - Research all **operating systems, software applications** and **data center equipment** operating **within** the **data center**
  - Review the **company's IT policies** and **procedures**
  - Evaluate the **company's IT budget** and **systems planning documentation**
  - Review the **data center's** **disaster recovery plan**

UNIX

- **Unix** is a multi-user, multi-tasking operating system.
- It's kernel's job to keep each process and user separate and to control access to system hardware, including cpu, memory, disk and other I/O devices.
- Linux is a free Unix-type operating system originally created by Linus Torvalds with the assistance of developers around the world.
- It originated in 1991 as a personal project of Linus Torvalds, a Finnish graduate student.

Linux distributions

- A Linux distribution is a collection of (usually open source) software on top of a Linux kernel. Examples:

- Ubuntu: http://www.Ubuntu.com/
- Kali Linux: http://www.kalilinux.com/
- Debian: http://www.debian.org/
- RedHat: http://www.redhat.com/
- Mandrake: http://www.mandrakesoft.com/
- Fedora: http://fedora.redhat.com/

Why linux/UNIX?

LINUX is free.

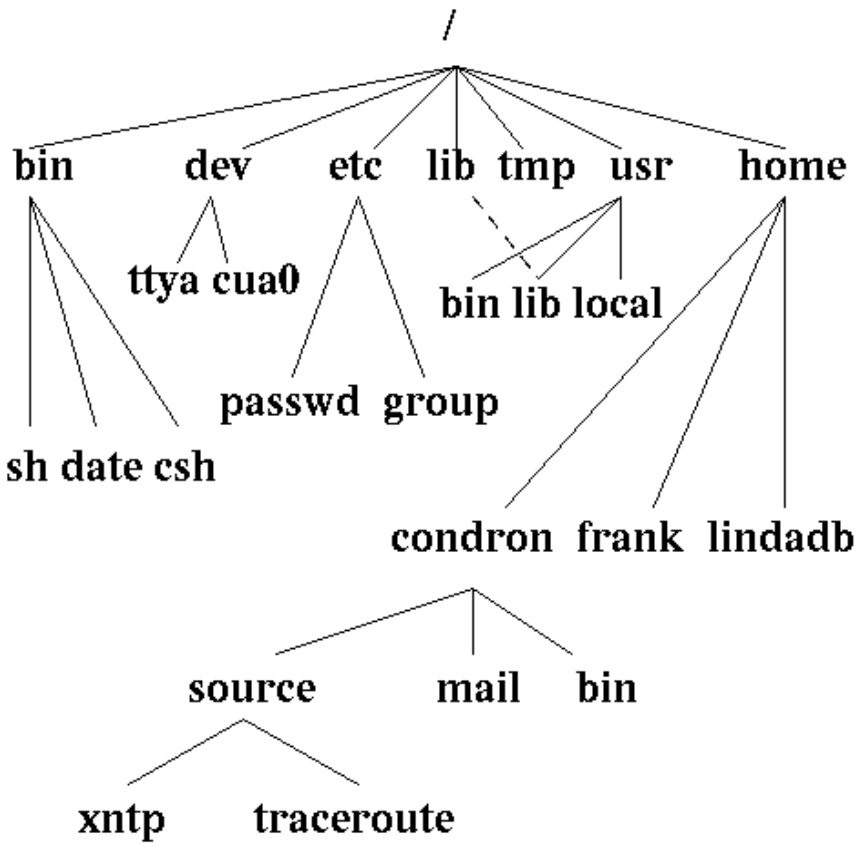
Can view and edit the source code of OS

- It is fully customizable.
- Most Important Feature is Stability

Important in shared environments and critical applications

- LINUX has better security structure.
- High Portability

LINUX/UNIX File System



File System

- Each node is either a file or a directory of files, where the latter can contain other files and directories.
- The full path name starts with the root, /, and follows the branches of the file system, each separated by /, until you reach the desired file, e.g.:
- /home/condron/source/xntp

Structure of Standard Directories in Unix/Linux

- / The **ancestor** of all directories on the system
- **/bin** Essential tools and other programs (or binaries).
- **/dev** Files representing the system's various hardware devices. For example, you use the file `/dev/cdrom' to access the CD–ROM drive.
- **/etc** Various system configuration files, startup files, etc.
- **/home** The home directories for all of the system's users.

Cont. ..

- **/lib** Essential system library files used by tools in `/bin'.
- **/proc** Files that give information about current system processes.
- **/root** The superuser's home directory, whose username is root.
- **/sbin** Essential system administrator tools, or system binaries.
- **/tmp** Temporary files.
- **/usr** Subdirectories with files related to user tools and applications.

Handling Man Pages/Help System

The **man** command which is short for manual provides in depth information about the requested command or allows users to search for commands related to a keyword. **Syntax**

$ man -k keywords

-k keywords: - shows list of man pages containing a string. Examples:

$man –k syslog - Lists help information on the syslog command.

$man mkdir - provides manual information about making directory command

Directories, Files and Inodes

- Every directory and file is listed in its parent directory.
- A directory is a file that contains a table listing the files contained within it, giving file names to the inode numbers in the list.
- The information about all the files and directories is maintained in INODE TABLE
- An Inode (Index Nodes) is an entry in the table containing information about a file (metadata) including file permissions, UID, GID, size, etc

File Commands

ls – directory listing
ls -al – formatted listing with hidden files
cd *dir* - change directory to *dir*
cd – change to home
pwd – show current directory
mkdir *dir* – create a directory *dir*
rm *file* – delete *file*
rm -r *dir* – delete directory *dir*
rm -f *file* – force remove *file*
rm -rf *dir* – force remove directory *dir* *
cp *file1 file2* – copy *file1* to *file2*
cp -r *dir1 dir2* – copy *dir1* to *dir2*; create *dir2* if it doesn't exist
mv *file1 file2* – rename or move *file1* to *file2*
if *file2* is an existing directory, moves *file1* into directory *file2*
ln -s *file link* – create symbolic link *link* to *file*
touch *file* – create or update *file*
cat > *file* – places standard input into *file*
more *file* – output the contents of *file*
head *file* – output the first 10 lines of *file*
tail *file* – output the last 10 lines of *file*
tail -f *file* – output the contents of *file* as it grows, starting with the last 10 lines

File Permissions

The system determines whether or not a user or group can access a file or program based on the permissions assigned to them.

One can easily view the permissions for a file by invoking a long format listing using the command ls -l.

The #'s can be:

0 = Nothing
1 = Execute
2 = Write
3 = Execute & Write (2 + 1)
4 = Read
5 = Execute & Read (4 + 1)
6 = Read & Write (4 + 2)
7 = Execute & Read & Write (4 + 2 + 1)

File Commands

chmod octal file– change the permissions of file to octal, which can be found separately for user(owner), group, and world(others) by adding:

- 4 – read (r)
- 2 – write (w)
- 1 – execute (x)

The three permissions are assigned to three defined categories of users. Examples:

chmod 777– read, write, execute for all

chmod 755– rwx for owner, rx for group and world

Process management

ps– display your currently active processes

pidof pro1 display process id named pro1

top– display all running processes

kill pid– kill process id pid

killall proc– kill all processes named proc*

bg– lists stopped or background jobs; resume a stopped job in the background

fg– brings the most recent job to foreground

fg n– brings job n to the foreground

Users & Groups Management

- Normally Linux computers have two user accounts—your own user account, and the root account, which is the super user that can access everything on the PC
- Ubuntu Linux uses groups to help you manage users, set permissions on those users.
- Linux stores a list of all users in the '/etc/groups' file

User Accounts storage

/etc/passwd This file contains the user account information for your system.

/etc/shadow This file contains passwords for your user accounts.

/etc/group This file contains your system's groups

Creating and Managing User Accounts

Using useradd

Syntax: **useradd** options username

E.g useradd mrx

You can also view these default values by entering useradd –D at the shell prompt.

Options

–c Includes the user's full name.

–e Specifies the date when the user account will be disabled. Format the date as yyyy-mm-dd.

–g Specifies the user's default group.

–m Specifies the user's home directory.

–p Specifies the user's password.

Cont ..

Using passwd

The passwd utility is used to change an existing user's password

You can find out this information using the –S option with passw

E.g passwd -S mrx

Syntax: passwd username

–l Locks the user's account.

–u Unlocks a user's account.

–d Removes a user's password.

Cont ..

Using usermod

From time to time, you will need to modify an existing user account.

Syntax: usermod *options username*

–*c* *Edits the user's full name.*

–*l* *Changes the username.*

–*l* *Locks the user's account.*

–*m* *Sets the user's home directory*

–*p* *Sets the user's password.*

Cont ..

- **Using userdel**

Syntax: **userdel** ***username***

ex: ***userdel*** ***mrx***

By default, userdel will not remove the user's home directory from the file system. If you do want to remove the home directory when you delete the user, you need to use the **–r** option in the command line. For example, entering **userdel –r mrx** will remove the account and delete his home directory.

Managing groups from the command line

Using groupadd

Syntax: **groupadd** *options* *groupname*

E.g groupadd admin_group

Options:

–**g** Specifies a GID for the new group.

–**p** Specifies a password for the group.

Cont ..

Using groupmod

To modify a group, including adding users to the group membership, you use the groupmod utility.

Syntax:

groupmod *options* *groupname*

Options:

–g Changes the group's GID number.

–p Changes the group's password.

Cont ..

Using groupdel

Syntax:

groupdel *group_name*

ex: ***groupdel*** **admin_group**

Supporting a Windows Network through SAMBA

Overview

A Samba file server enables file sharing across different operating systems over a network. It lets you access your desktop files from a laptop and share files with Windows and Linux users.

Installing Samba

To install Samba, we run:

sudo apt update

sudo apt install samba

We can check if the installation was successful by running:

whereis samba

Setting up Samba

Now that Samba is installed, we need to create a directory for it to share:

mkdir /home/<username>/sambashare/

The configuration file for Samba is located at /etc/samba/smb.conf. To add the new directory as a share, we edit the file by running:

sudo nano /etc/samba/smb.conf

Setting up Samba ...

At the bottom of the file, add the following lines:

[sambashare]

comment = Samba on Ubuntu

path = /home/username/sambashare

read only = no

browsable = yes

Then press Ctrl-O to save and Ctrl-X to exit from the nano text editor.

Setting up Samba ...

Now that we have our new share configured, save it and restart Samba for it to take effect:

sudo service smbd restart

Update the firewall rules to allow Samba traffic:

sudo ufw allow samba

Setting up User Accounts and Connecting to Share

Since Samba doesn't use the system account password, we need to set up a Samba password for our user account:

sudo smbpasswd -a username

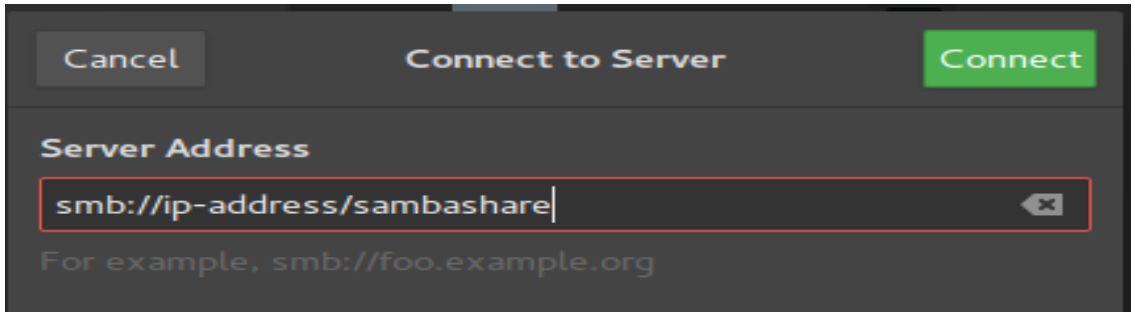
Note

Username used must belong to a system account, else it won't save.

Connecting to Share

On Ubuntu: Open up the default file manager and click Connect to Server then enter:

Setting up User Accounts and Connecting to Share…



On Windows, open up File Manager and edit the file path to:

\\ip-address\sambashare

Note: ip-address is the Samba server IP address and sambashare is the name of the share.

Thanks